

BEYOND THE ORIGINAL

How Product Variations Influence Purchase Intent, Customer Clusters, & Cannibalization Risk

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IMC 498



BUSINESS CONTEXT



- Launched in 2013, the **original** Bartholomew Bear has been a cornerstone of Jellycat's product line.
- Since 2023, Jellycat has **expanded** the Bartholomew Bear collection with 9 variations, *aiming to capitalize on the character's popularity and broaden appeal.*

*Source: <https://us.jellycat.com/bartholomew-bear/>

BUSINESS PROBLEM

Do variations of Bartholomew Bear *increases* potential buyers or risks *cannibalization* of the original Bear?





Main Research Question:

How does purchase intent for the original Bartholomew Bear change after variations are introduced?

Sub-Questions:

Q1: Are there distinct customer clusters based on variation preferences?

Q2: Can we identify customer-variation clusters and audience duplication across products?

Q3: Do variations reduce the original bear's appeal (cannibalization risk)?

SURVEY DESIGN & DATA COLLECTION

Target Audience

Full-time IMC graduate students at Northwestern University (**N = 127**)

Response Rate

42 out of **127** students responded
≈33% response rate

Data Collection Tool

Google Forms survey (anonymous & self-administered)

Collection Timeline

5 Days

Design Structure

- Closed-ended & image-based questions
- Multi-select & single-select formats
- Attribute prioritization to reveal key purchase drivers
- Demographics (age, gender) collected for segmentation insights
- Substitution logic tested to follow-up on original bear choice



Pick your favorite Bartholomews! 🐻

Hi IMC friends! Please fill out this quick survey for our research project. This survey will take less than one minute. Thank you!! :))

* Indicates required question

Which of the following Bartholomew Bear variations would you consider purchasing for yourself? (Select all that apply) *



☐ Bathrobe Bartholomew



☐ Eggplant Bartholomew



☐ Heartthrob Bartholomew



☐ Bat Bartholomew



If the variations above were available, would you still consider purchasing the original Bartholomew Bear for yourself? *



- ☐ Yes
- ☐ No
- ☐ I already owned the original Bartholomew Bear since before 2023.

Which of the following attributes **most strongly** determines your decision to purchase a Bartholomew Bear? *

- ☐ Huggability
- ☐ Collectability
- ☐ Cuteness
- ☐ Nostalgia
- ☐ Uniqueness

Gender *

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

Age *

Your answer

Submit

Clear form





Q1.

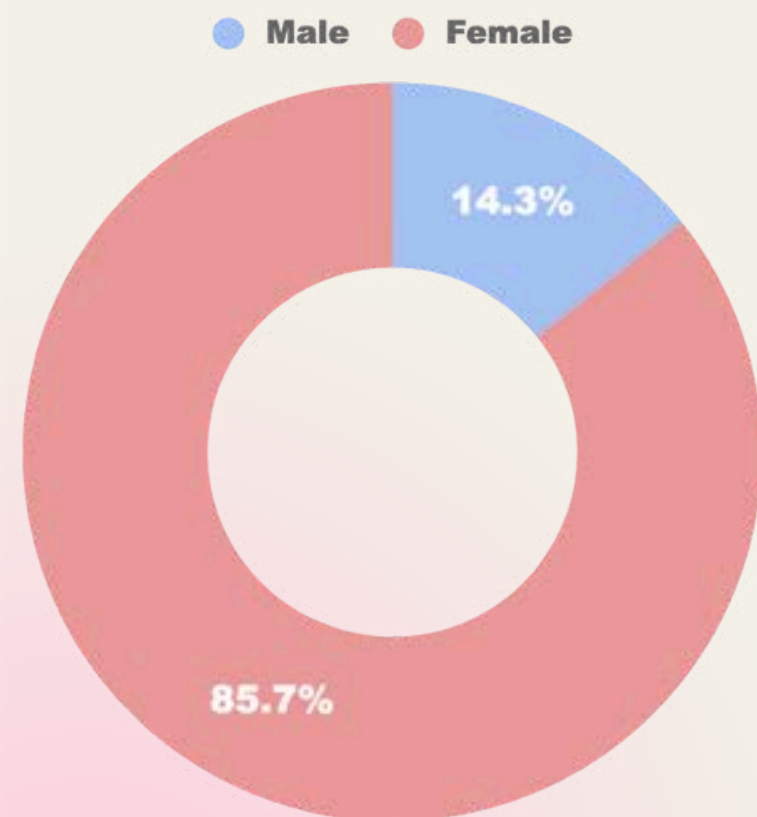
Are there **distinct** customer clusters based on **variation preferences**?





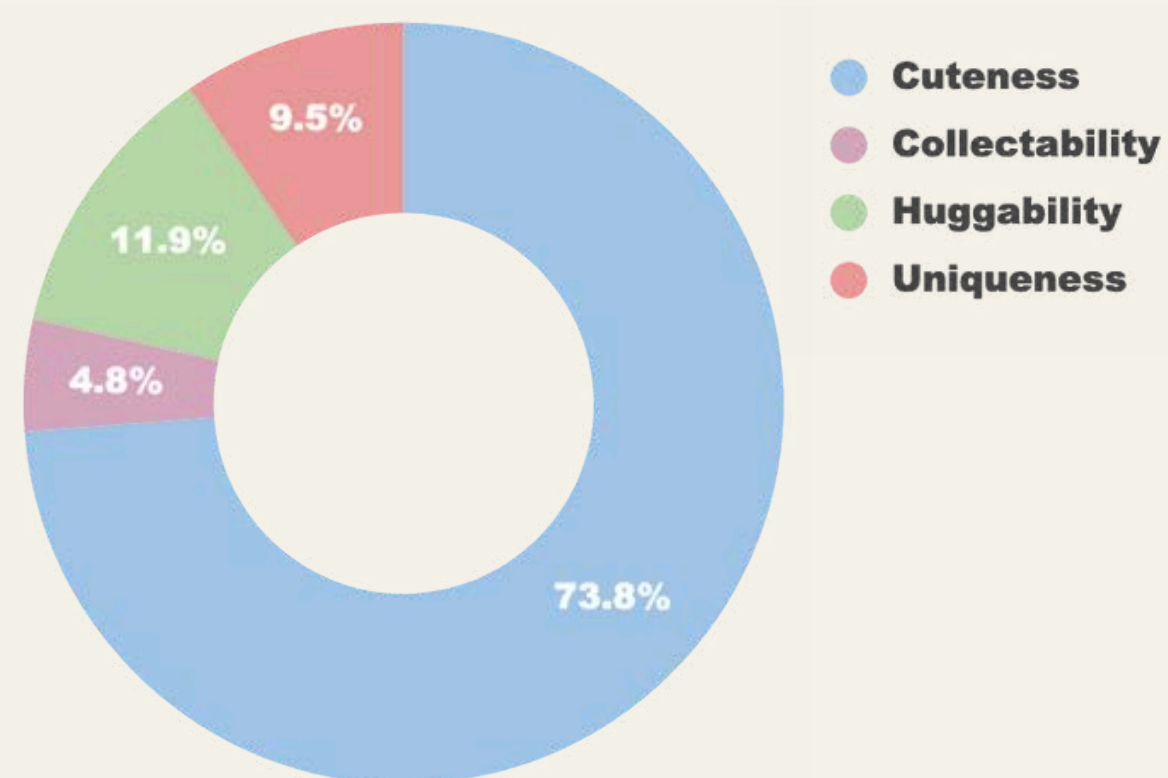
SAMPLE CUSTOMER PROFILE

Gender & Age



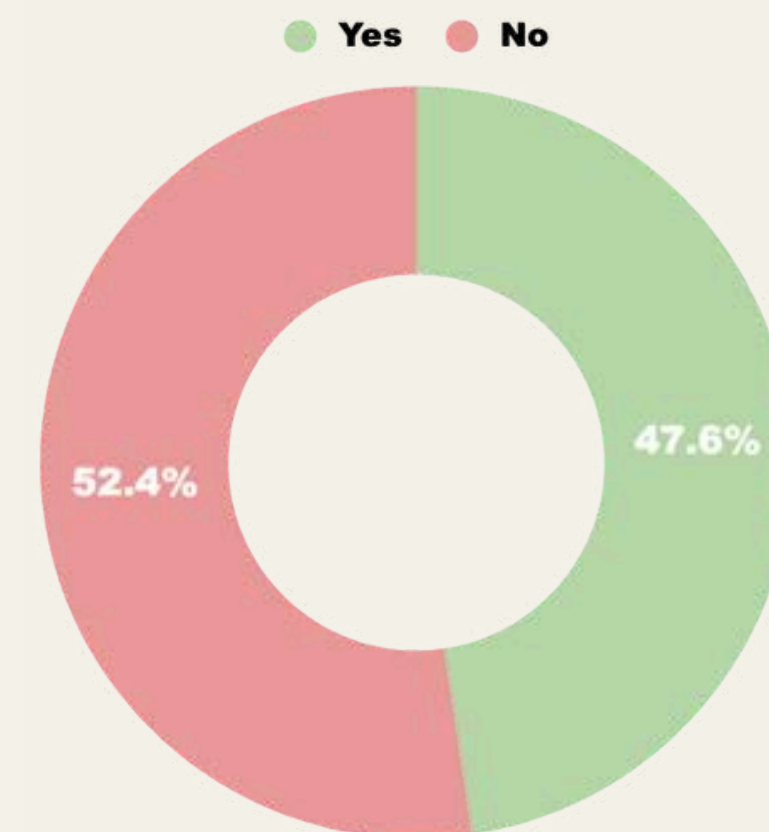
Predominantly **female** with a mean age of **24**

Product Features



Cuteness is the primary customer preference

OG Bear Purchase Intent



Relatively **balanced** distribution regarding purchase intent

CUSTOMER ONE-MODE ANALYSIS

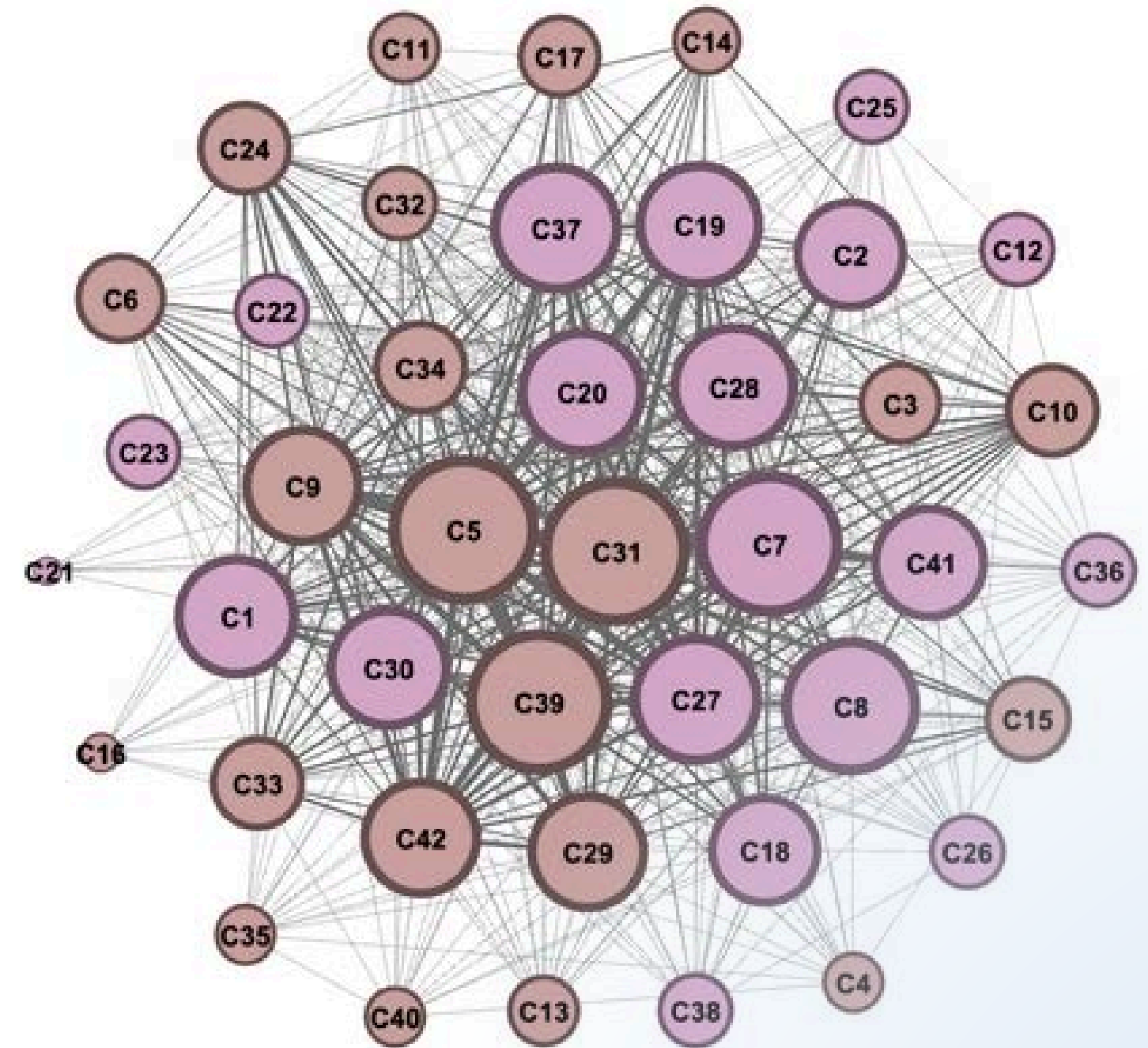
Average Degree: **27**

Clustering Coefficient: **0.855**

Degree centrality ranges from 8 to 41, indicating varying levels of product preference overlap among customers.

→ **High-degree users reflect broad product preferences**, connecting with nearly all others, bridging segments, & strengthening network cohesion

→ **Low-degree users reflect niche preferences**, engaging with fewer peers & having less influence on collective trends



COMMUNITY DETECTION & SEGMENTATION ANALYSIS

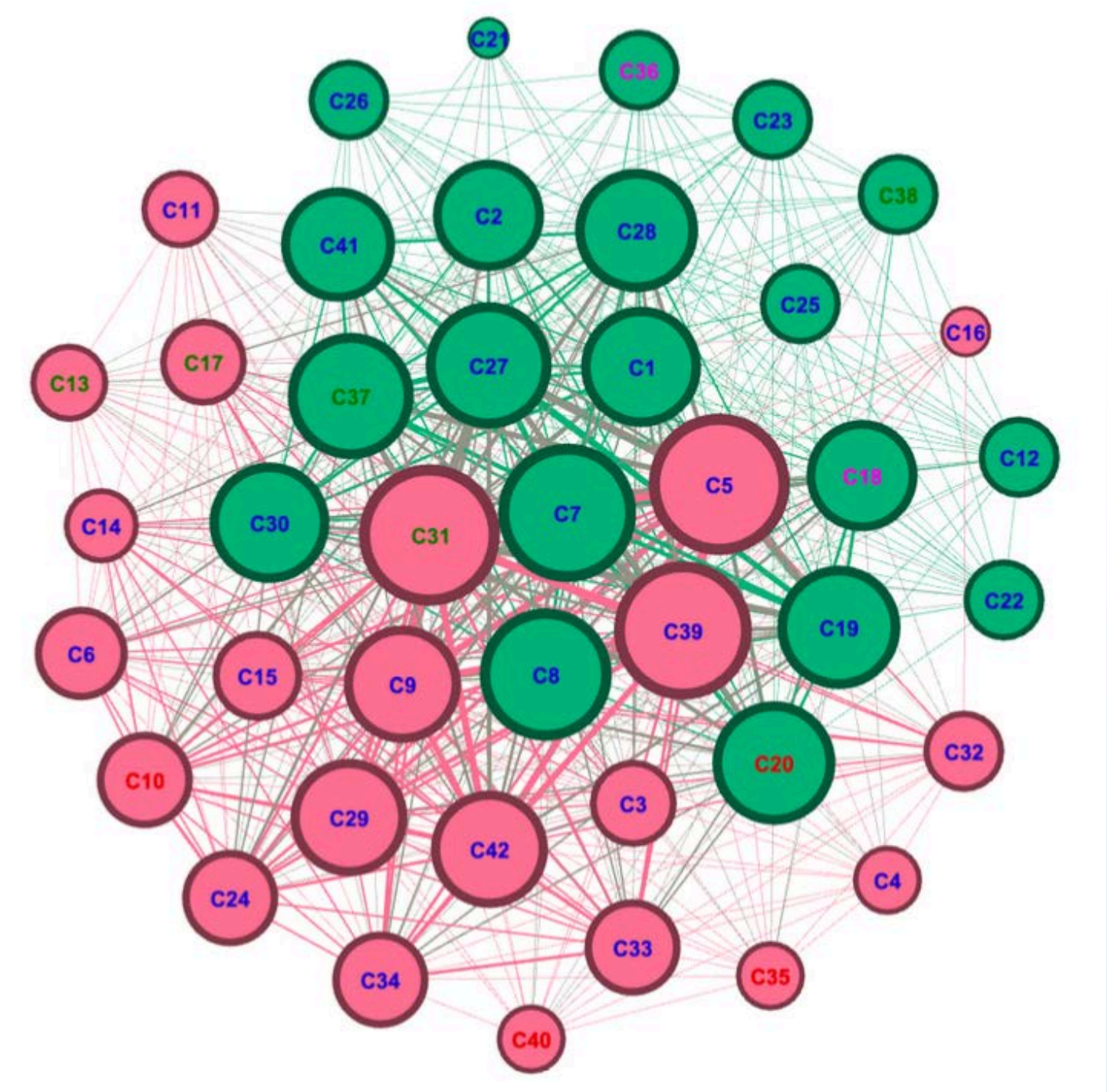
Modularity Score: **0.106** ; *Weakly defined clusters*

Clusters Detected: **2**

Class	# Customers	Values				OG Bear	
		Cute	Hug	Unique	Collect	Yes	No
1	22	17	2	3	0	12	10
2	20	15	2	1	2	8	12

Cuteness is the primary product feature preference for both Cluster 1 and Cluster 2.

- **Cluster 1** values *uniqueness* slightly more and has more customers who intend to purchase the OG bear.
- **Cluster 2** contains all *collectors*, indicating a *niche collector subgroup* and has more customers who do not intend to purchase the OG bear.





Q2.

Can we identify **customer-variation clusters** and **audience duplication** across products?



PRODUCT PREFERENCE OVERLAP

Variations with Highest Willingness to Purchase

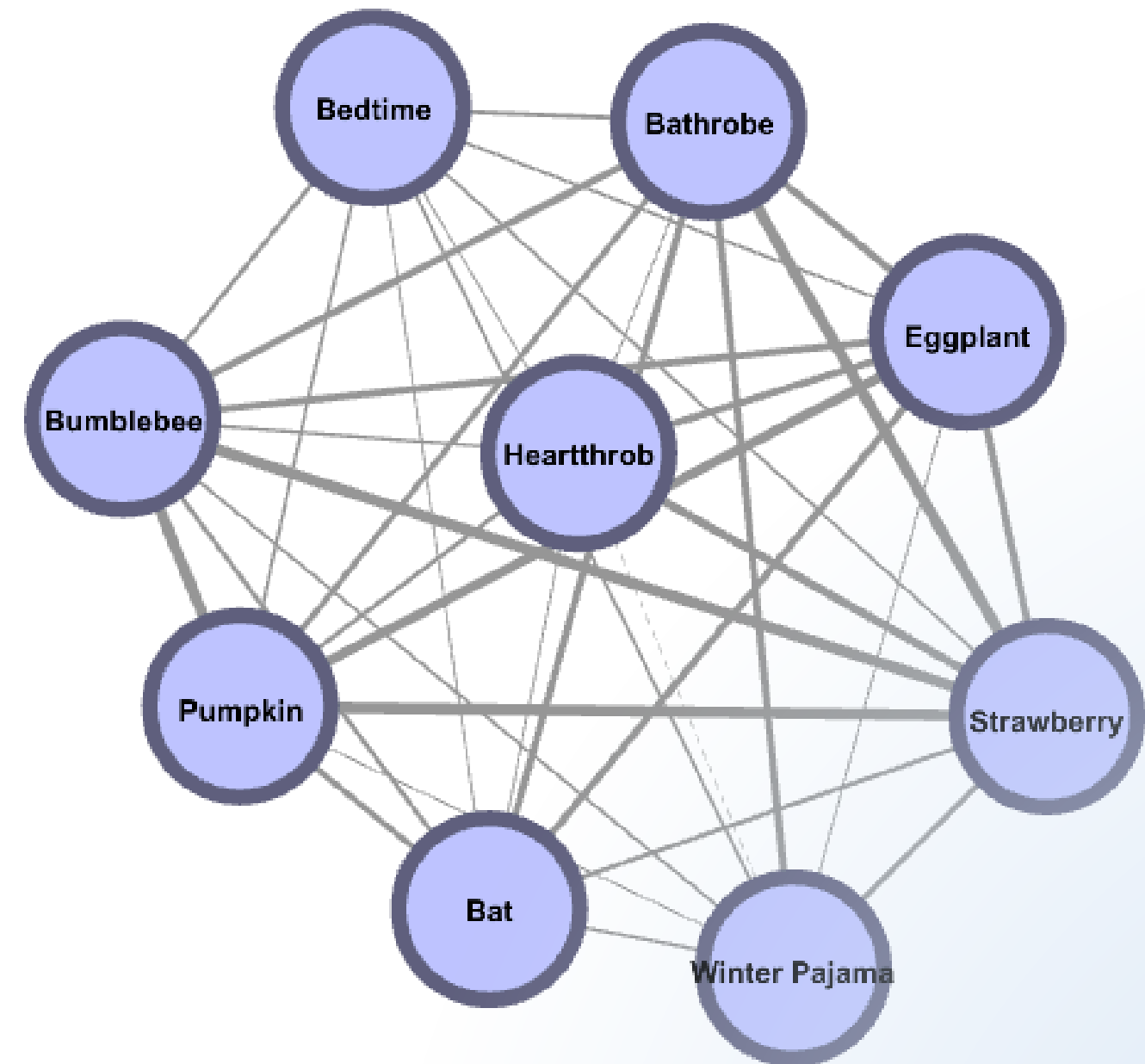
- Bathrobe (22)
- Strawberry(21)
- Eggplant(18)



High Co-Selection Nodes (Central Hubs)

No distinct product clusters exist and users browse broadly across categories, not within isolated groups.

Strawberry and *Bumblebee* also show **high connectivity**, making them strong secondary anchors.



PRODUCT PREFERENCE OVERLAP

Three Best Pairings

- Pumpkin and Bumblebee
- Strawberry and Bumblebee
- Strawberry and Pumpkin

*Can be bundled, co-promoted
or used as entry points in
recommendation engines.*

	Bumblebee			Strawberry			Pumpkin		
	Shared Audience	%V	%H	Shared Audience	%V	%H	Shared Audience	%V	%H
Bumblebee	15			11	52%	73%	11	69%	73%
Strawberry	11	73%	52%	21			11	69%	52%
Pumpkin	11	73%	69%	11	52%	69%	16		
OG	9	60%	45%	12	57%	60%	10	63%	50%

See Appendix for the whole table

PRODUCT PREFERENCE OVERLAP



Audience Duplication with OG Bear

Source	Yes	No	%Yes
Bathrobe	10	12	45%
Eggplant	10	8	56%
Heartthrob	7	5	58%
Bat	7	2	78%
Bumblebee	9	6	60%
Strawberry	12	9	57%
Winter Pajama	5	4	56%
Bedtime	6	2	75%
Pumpkin	10	6	63%

- **Bat & Bedtime:** *Highest* audience duplication with OG bear (**78%/75%** who chose Bat/Bedtime bears intent to purchase OG bear)
- **Bathrobe:** *Lowest* audience duplication with OG bear (i.e., **45%** of customers who chose Bathrobe intent to purchase OG bear)

See Appendix for the whole table



Q3.

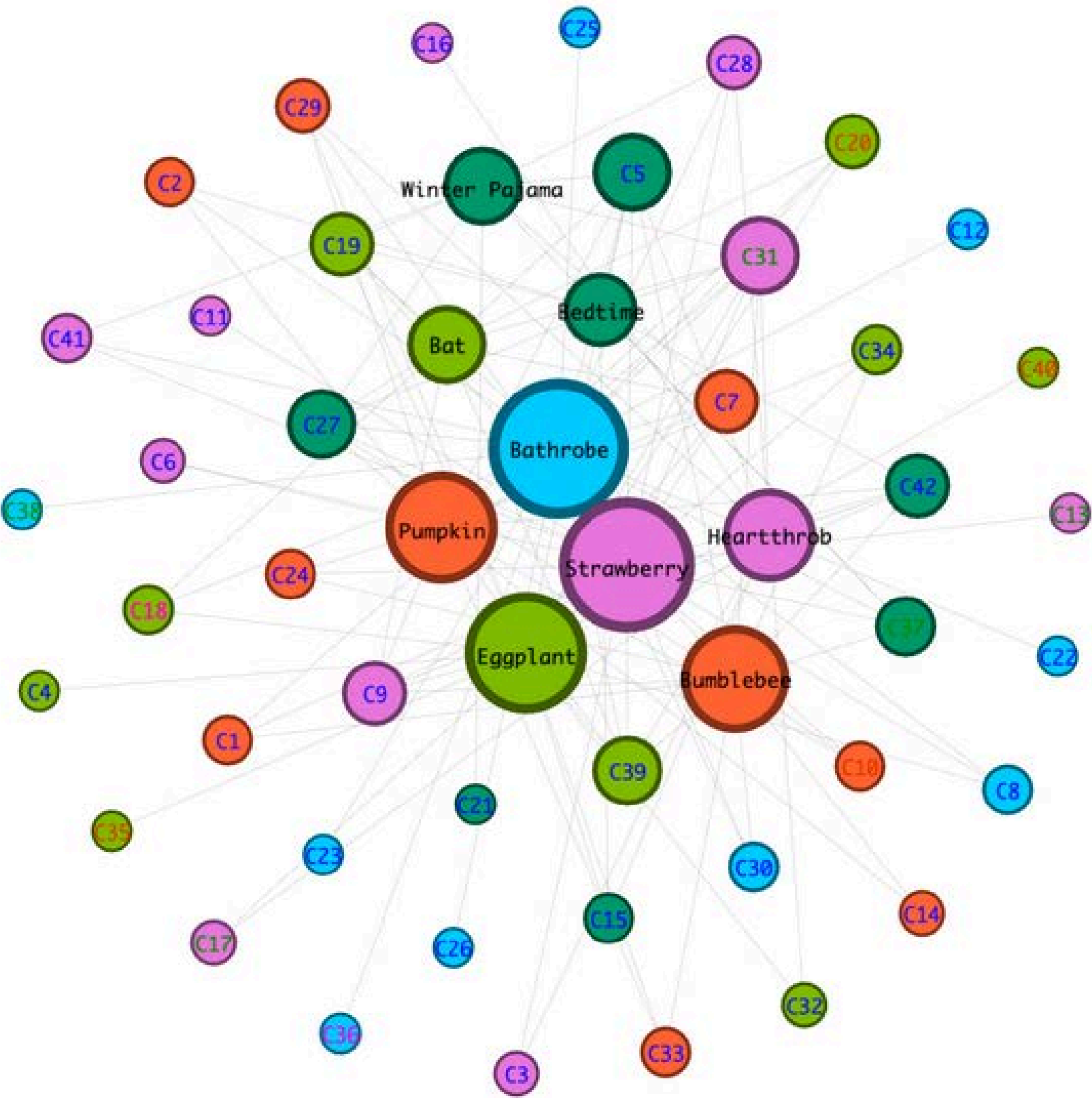
*Do variations **reduce** the original bear's appeal (**cannibalization risk**)?*



TWO-MODE ANALYSIS

- Modularity Score: **0.239** (Res = 1, **5 Clusters**)
- Node Label Color (Below):
 - Blue – Cuteness
 - Red – Uniqueness
 - Green – Huggability
 - Pink – Collectability

Class	Product Description	#Customer	Values				OG Bear Intent	
			Cute	Hug	Unique	Collect	Yes	No
1	Bathrobe	9	7	1	0	1	1	8
2	Strawberry & Heartthrob	10	7	3	0	0	5	5
3	Eggplant & Bat	9	5	0	3	1	5	4
4	Bumblebee & Pumpkin	8	7	0	1	0	4	4
5	Winter Pajama & Bedtime	6	5	1	0	0	5	1



VARIATION-CUSTOMER CLUSTERS



CLUSTER #1: BATHROBE

- Primary product feature: cuteness > huggability & collectability
- Has the **highest** # and % of customers who **don't intent** to purchase OG bear



CLUSTER #2: STRAWBERRY & HEARTTHROB

- Primary product feature: cuteness > huggability
- Customers who prefer huggability concentrate in this cluster
- Has a **balanced** purchase intent for OG bear



CLUSTER #3: EGGPLANT & BAT

- Primary product feature: cuteness > uniqueness > collectability
- Customers who prefer uniqueness concentrate in this cluster
- Most diverse customer preferences
- Has a **balanced** purchase intent for OG bear

VARIATION-CUSTOMER CLUSTERS



CLUSTER #4: PUMPKIN & BUMBLEBEE

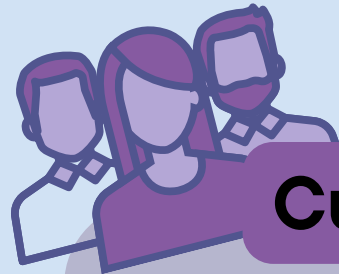
- Primary product feature: **cuteness** > **uniqueness**
- Has a **balanced** purchase intent for OG bear



CLUSTER #5: WINTER PAJAMA & BEDTIME

- Primary product feature: **cuteness** > huggability
- Has the **highest** # and % of customers who **do intent** to purchase OG bear

CONCLUSIONS



Customer Clustering

Cluster 1 (Uniqueness-focused):
More likely to purchase OG Bear
and prefer distinct variations.

Cluster 2 (Collectability-focused):
Less likely to purchase OG Bear
and prefers collectible variations

**C2 poses a higher
cannibalization risk.**

Product Audience Duplication

Bathrobe

- Highest degree centrality
- Lowest duplication with OG Bear
- **High cannibalization risk.**

Bat & Bedtime

- Highest overlap with OG Bear
- Strong potential for **cross-promotion**



Two-Mode Clusters



Bathrobe Cluster

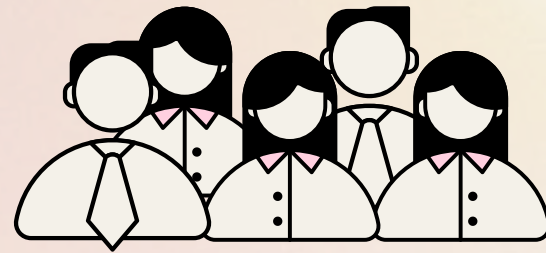
Isolated and does not overlap with
other products
High cannibalization risk.

Winter PJ & Bedtime Cluster

Strong OG Bear affinity
Low risk of cannibalization.

BUSINESS IMPLICATIONS

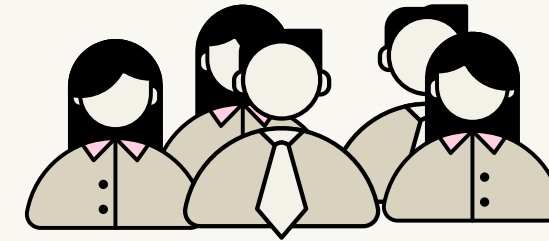
CUSTOMER SEGMENT



CLUSTER 1

- Higher willingness to purchase OG bear
- Value uniqueness more

**Emphasize on OG bear's
uniqueness in marketing narrative**



CLUSTER 2

- Lower intent of purchasing OG bear
- Value collectability more

**Minimize the marketing effort of
OG bear.**

BUSINESS IMPLICATIONS

PRODUCT



Leverage the "Triangle"

Strawberry, Bumblebee, and Pumpkin

High-affinity bundle in
personalized recommendations



Avoid OG Bear + Bathrobe

Bathrobe will dilute consumer's
willingness to purchase OG Bear



Pair OG Bear with Bat, Bedtime or Winter Pajama

Tap into stronger preference
overlap and enhance conversion
potential.



Thank you!

APPENDIX

PLEASE FIND THE LINK BELOW FOR:

SURVEY RESPONSES

AUDIENCE DUPLICATION TABLE

TWO-MODE MATRIX

PRODUCT ONE-MODE MATRIX

CUSTOMER ONE-MODE MATRIX

LINK HERE

Timestamp	Score	Which of the following Bartholomew Bear variations would you consider purchasing for yourself? (Select all that apply)	If the va
5/23/2025 12:24:57		Bathrobe Bartholomew, Bumblebee Bartholomew, Strawberry Bartholomew	Yes
5/23/2025 12:29:46		Bathrobe Bartholomew, Bedtime Bartholomew, Pumpkin Bartholomew	Yes
5/23/2025 12:32:19		Heartthrob Bartholomew, Strawberry Bartholomew	No
5/23/2025 12:33:23		Eggplant Bartholomew	Yes
5/23/2025 12:34:00		Bathrobe Bartholomew, Eggplant Bartholomew, Heartthrob Bartholomew, Bat Bartholomew, Bumblebee Bartholomew, Strawberry Bartholomew, Winter Pajama Bartholomew, Bedtime Bartholomew, Pumpkin Bartholomew	Yes
5/23/2025 12:34:00		Strawberry Bartholomew, Pumpkin Bartholomew	No
5/23/2025 12:34:01		Bathrobe Bartholomew, Eggplant Bartholomew, Bat Bartholomew, Strawberry Bartholomew, Winter Pajama Bartholomew, Pumpkin Bartholomew	No
5/23/2025 12:34:14		Bathrobe Bartholomew, Eggplant Bartholomew, Strawberry Bartholomew	No
		Eggplant Bartholomew, Heartthrob	
Form Responses 1 Audience Duplication Customer Nodes User-Product User-User Product-Product > <			